

PROBLEMS CLASS 4

In this final problems class of term, we will look at two old exam questions which model medical applications. The first considers how quickly wounds heal themselves; and the second looks at vaccine efficacy, marking five years since the announcement of the efficacy of the first Covid vaccines.

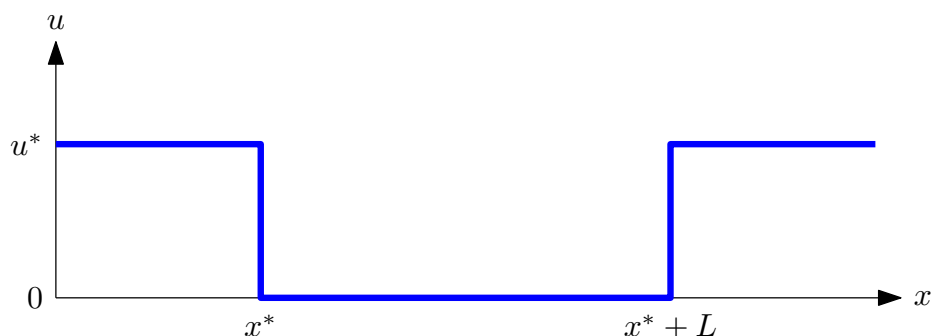
1. Travelling waves in skin repair (Exam 2022, section B)

A model of wound healing in one spatial dimension is given by

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - u \log \left(\frac{u+a}{K} \right), \quad (1)$$

where $u(x, t)$ is the density of healthy tissue at position x and time t ; and where a and K are positive constants, with $0 < a < K$.

- Consider first the homogeneous form of (1). Find the equilibria and determine their stability in this case.
- Describe qualitatively the full model, (1). In what way is it different to that described by the Fisher–Kolmogorov equation?
- We are going to look for travelling wave solutions, i.e. solutions of the form $u(z)$, where $z = x - ct$ and $c > 0$. Given the stability of the equilibria in the homogeneous system, suggest suitable boundary conditions for $u(z)$ at $z = \pm\infty$, explaining your choice.
- Now write (1) for $u(z)$, and by performing a linear stability analysis, determine the stability of the homogeneous equilibria. Given that waves travel at their lowest possible speed, determine the wavespeed, c .
- Consider now a wound which is initially of length L , i.e. as in the diagram below, where u^* is the value of the nonzero homogeneous equilibrium of (1):



Illustrating your answer with sketches of $u(x, t)$ for increasing t , explain what you expect to happen for subsequent times and determine approximately how long it will take for the wound to heal.

2. Disease modelling with vaccination (Exam 2022, section B)

Parliament, 10 November 2020 (Hansard):

Jonathan Ashworth, shadow secretary of state for health and social care: What is the government's current working assumption of the proportion of the population that needs to be vaccinated to establish herd immunity and bring R below 1?

Matt Hancock, secretary of state for health and social care: The honest truth to that question is that we do not know what proportion of the population the vaccination needs to reach in order for it to stop the epidemic.

Consider the following extension of the SIR model for an infectious disease, where susceptible individuals are vaccinated at a rate ν ,

$$\frac{dS}{dt} = \alpha R - \beta IS - \nu S, \quad (2)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (3)$$

$$\frac{dR}{dt} = \gamma I - \alpha R + \nu S,$$

where $\alpha, \beta, \gamma, \nu$ are all positive constants, and $S(t), I(t), R(t) \geq 0$ to be permissible.

- (a) Show that $N = S + I + R$ is constant and use this to eliminate R from (2). Explain why, from now on, we only need to consider the subsystem formed of equations (2) and (3).
- (b) Give an interpretation of the constants α, β, γ, N . What would be suitable units for these constants?
- (c) It is proposed that this could be an appropriate model for the spread of Covid-19. Is this a suitable model for this problem? Give three clear arguments, which can be in favour of this as a suitable model, or against its suitability. Your three arguments do not have to argue the same way.
- (d)
 - (i) Find the values of S and I at the two equilibria of the system.
 - (ii) The disease is eradicated if the disease-free equilibrium is the only possible stable equilibrium and the endemic equilibrium does not exist. Perform a linear stability analysis on both equilibria of the system and show that the disease is eradicated if $R^* < 1$, where you should define the constant R^* .
 - (iii) Define R_0 as the value of R^* in the absence of vaccination (i.e. when $\nu = 0$): in this way, R_0 can be seen as an inherent property of the disease. Rearrange the statement $R^* < 1$ to find the condition on the vaccination rate, ν , for the disease to be eradicated in terms of R_0 .
- (e) You are now asked to present the same argument graphically. Draw a phase portrait of I against S and, by moving the nullclines on your portrait, show how the existence of the endemic equilibrium depends on the vaccination rate, ν , and the value of R_0 . You do not need to explicitly draw trajectories to make this argument.